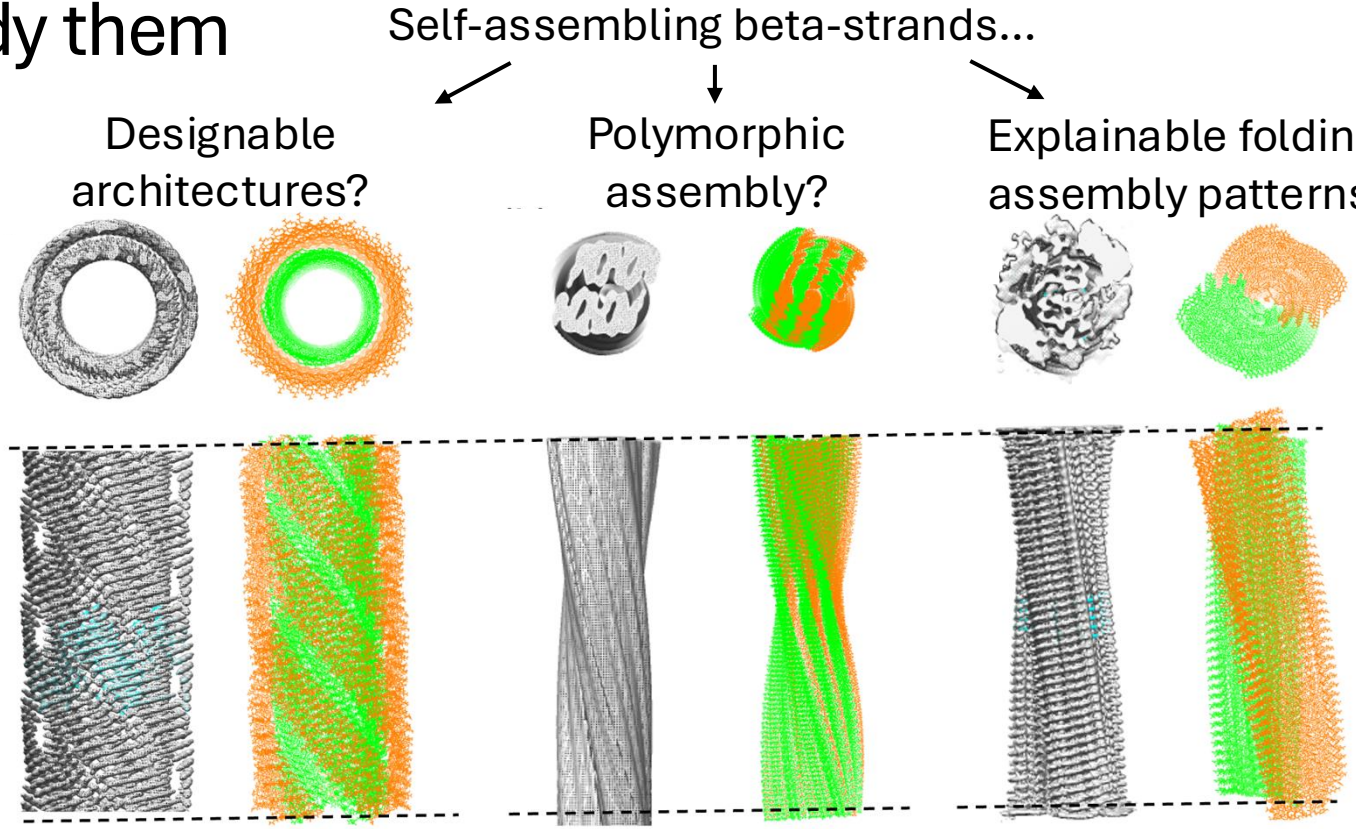
2026 Building Molecular Systems Workshop

FibrilGen: a Python package to generate cross-beta nanostructures

Speaker: Chao-Yu Yang

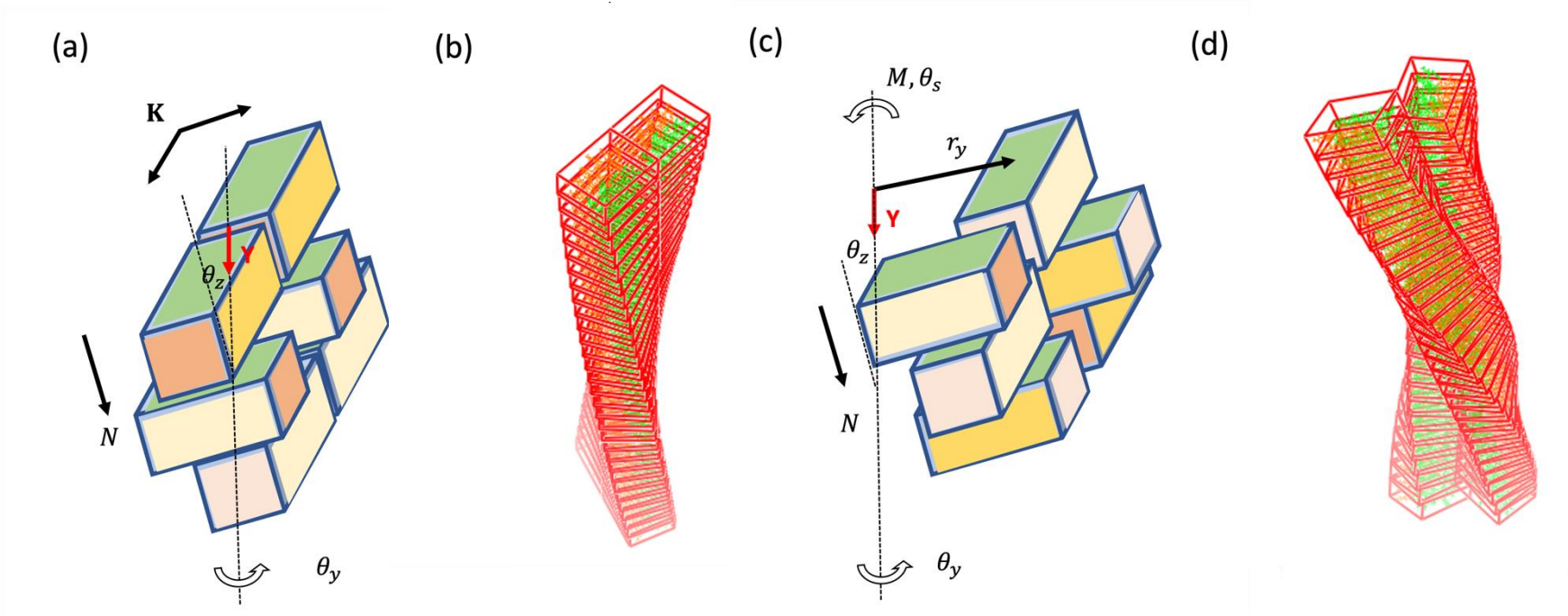
Date: 27/11/2026

What are cross-beta nanostructures and why we want to study them

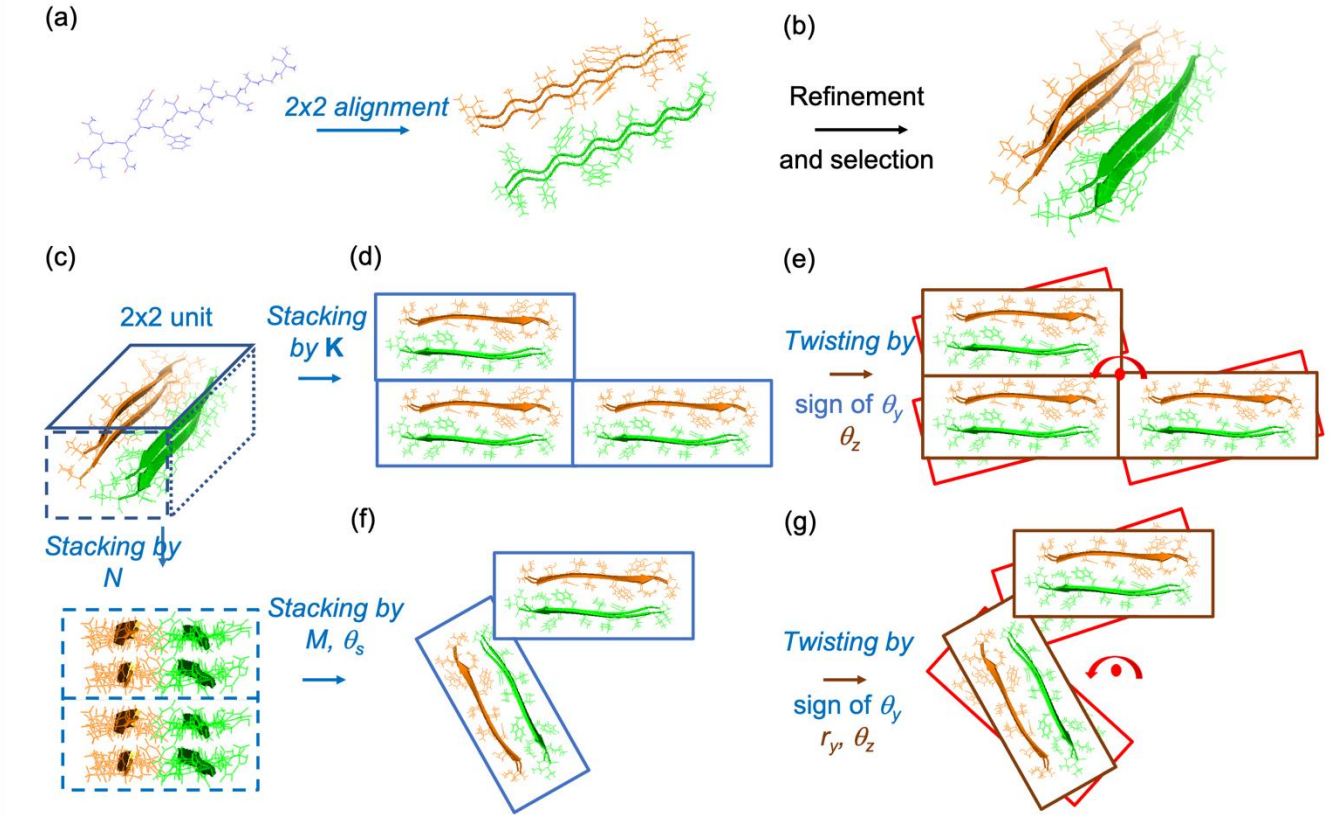


Wang, F. et al. (2021), Schmidt, A. et al. (2016), Gremer, L. et al. (2017), Yang, C.-Y. et al. (2025)

An overview of common cross-beta geometry

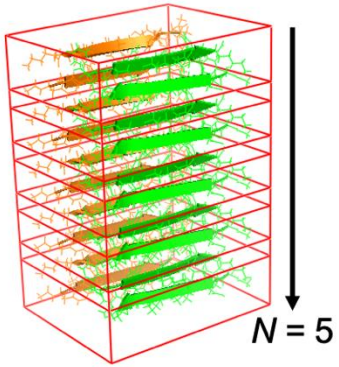


Assemble a peptide molecule into a cross-beta structure



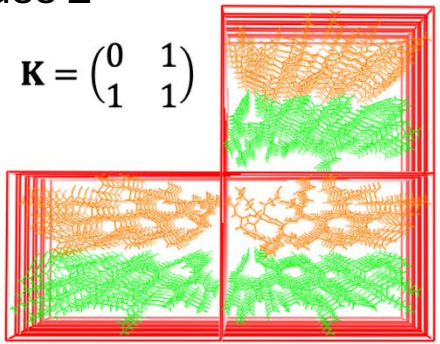
Seven parameters for six classes of cross-beta models

Class 1



Class 2

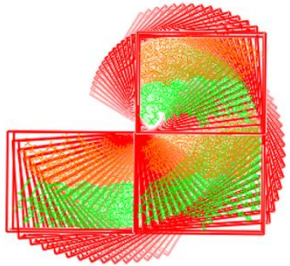
$$\mathbf{K} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$$



Class 3



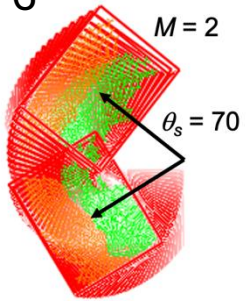
Class 4



Class 5



Class 6



Let us build cross-beta structures!

Peptide, periodic building blocks, nanofibrils and more …

Setup a Colab working environment

- Colab file: https://github.com/ChaoYuYang0/FibrilGen-v1/blob/main/demo_Colab.ipynb

Setup a Python environment

First of all, we setup an environment with essential Python libraries.

```
[ ]
!pip install pymol-open-source==3.2.0a0
!pip install openbabel==3.2.1
!pip install py3Dmol==2.5.5
!git clone https://github.com/ChaoYuYang0/FibrilGen-v1
```

Change the working directory

Once we finish the setup, we change the working directory to "FibrilGen-v1".

```
[ ]
%cd FibrilGen-v1/
```



Build a beta-strand

Basic

Input: a peptide sequence (one-letter code format)

Output: a beta-strand (PDB format)

Advanced

How do side–chain rotations create steric clashes?

Interface

Step 1: build your beta-strand

In this section, we change the input peptide sequence and visualise the output peptide molecule. We change the side-chain rotation angles to see how they can introduce steric clashes (optional).

[ ]

Peptide sequence

Side-chain rotation angles

Input_seq

FEFKFEFK

Angle_sidechain

1_n_ca_cb_cg_-20,3_n_ca_cb_cg_-20,5_n_ca_cb_cg_-20,7_n_ca_cb_cg_-20

Ouput_filename

F8.pdb

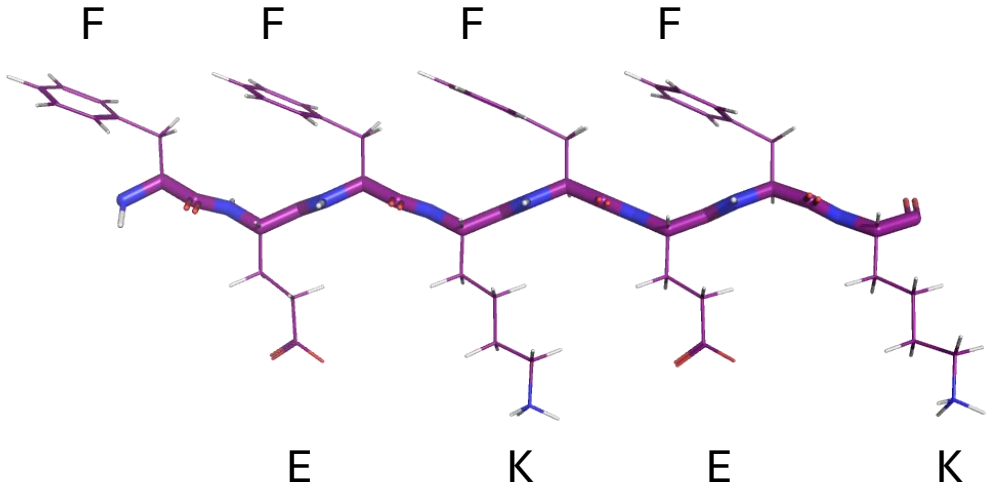
Show_the_structure

Show code

Side-chain rotation angles written as: residx_atom1_atom2_atom3_atom4_angle1, ...

Practice

1. Run the example
2. Try a different peptide sequence
3. Try a different side-chain rotation angle





Build a bilayer structure

Basic

Input: a peptide molecule (PDB format), peptide alignment for a 2 x 2 unit

Output: an energy minimised beta-sheet (PDB format)

Advanced

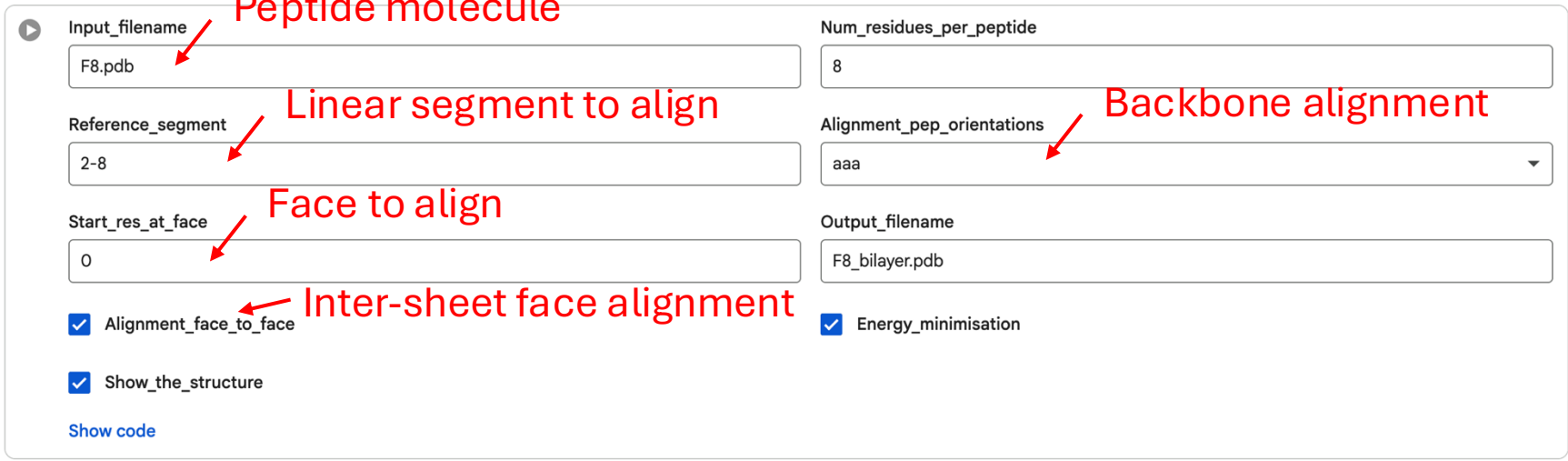
How peptide molecule and peptide alignment destabilise the beta-sheet assembly?

Interface

Step 2: Build your bilayer structure

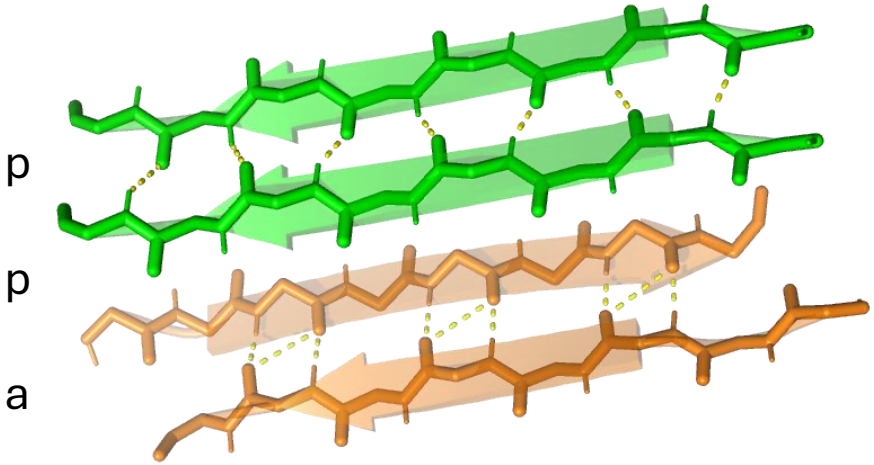
In this section, we change the input peptide alignment and visualise the output 2 x 2 beta-sheet building block. We multiply the 2 x 2 building block 5 times, then optimise the assembled structure. We check whether inter-peptide backbone hydrogen bonds and side-chain packing are plausible in the assembled structure.

[ ]

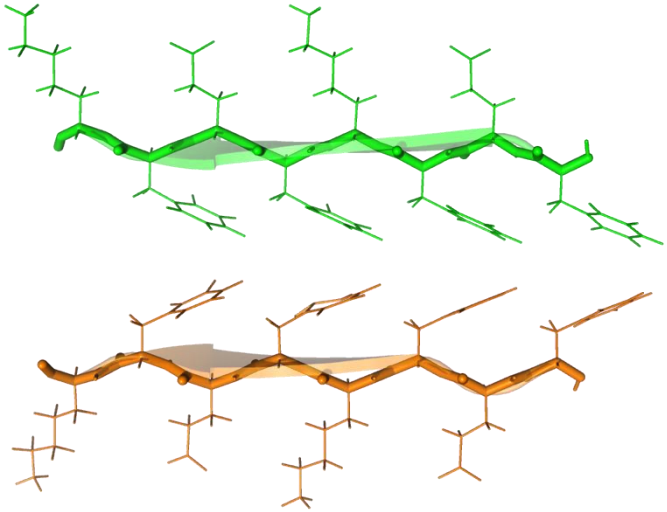


Practice

1. Run the example
2. Try a different peptide molecule
3. Try a different peptide alignment



Aligned face-to-face





Build a fibril structure

Basic

Input: reference bilayer (PDB format), input parameters for a fibril model

Output: a cross-beta structure (PDB format)

Advanced

How do 2 x 2 units affect the feasible assembly of beta-sheet structure?

Interface (class 1 **sheet** structures)

Step 3: Build your fibril structure

In this section, we use a 2 x 2 building block to assemble different classes of cross-beta structures.

(a) We use our optimised bilayer to assemble a **sheet** structure. We change the number of stacking (parameter N) along the beta-sheet axis and visualise the structure.

[ ]

Reference bilayer

Input_filename: F8_bilayer.pdb

Input_num_peptides_per_sheet: 10

Num_residues_per_peptide: 8

Reference_segment: 2-8

FibrilGen parameters

N: 10

Output_filename: F8_sheet.pdb

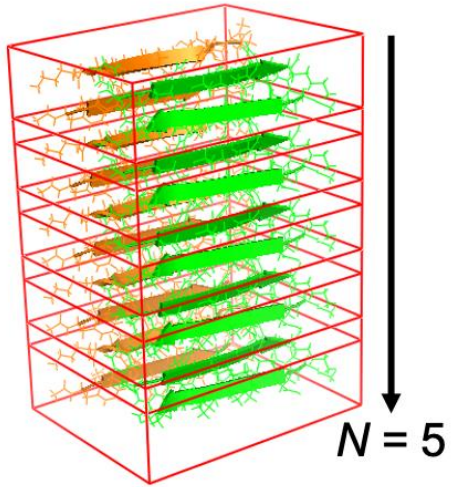
Show_the_structure

- Parameter N is an integer repeating stacking along the vertical axis.

Show code

Practice (class 1 **sheet** structures)

1. Run the example
2. Try a different *N*. Increase the number for a longer fibril.



Interface (class 2 **stacked sheet** structures)

(b) We use our optimised bilayer to assemble a **stacked sheet** structure. We change the number of stacking on the sheet cross-section (parameter **K**, do not specify element 1 surrounded by element 0) and the number of stacking (parameter N) along the beta-sheet axis.

[ ]

Reference bilayer

Input_filename: F8_bilayer.pdb

Input_num_peptides_per_sheet: 10

Num_residues_per_peptide: 8

Reference_segment: 2-8

FibrilGen parameters

N: 10

K: 0_1_0,1_1_1

Output_filename: F8_stacked_sheet.pdb

Show_the_structure

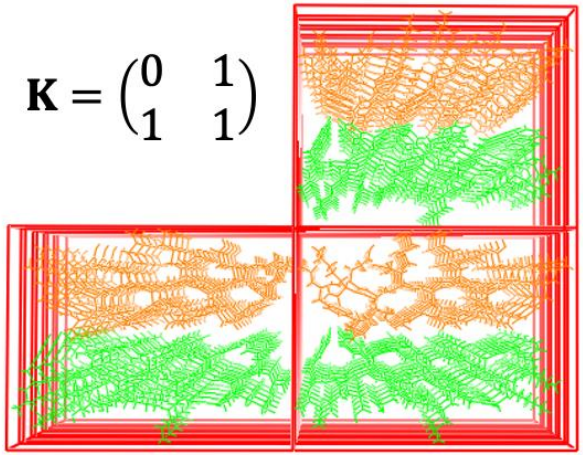
- Parameter N is an integer repeating stacking along the vertical axis.
- Parameter **K** is a binary matrix ("_" to seperate columns, "," to sperate rows) assigning whether a 2 x 2 unit is presented.

Show code

Practice (class 2 **stacked sheet** structures)

1. Run the example
2. Try a different *N*. Increase the number for a longer fibril.
3. Try a different **K**. Increase the number of presented units for a thicker fibril.

$$\mathbf{K} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$$



Interface (class 3 **rod** structures)

(c) We use our optimised bilayer to assemble a **rod** structure. We change the twist chirality (the sign of parameter θ_y), the initial tilt angle (parameter θ_z), and the number of stacking (parameter N) along the fibril axis.

[ ]

Reference bilayer

Input_filename: F8_bilayer.pdb

Input_num_peptides_per_sheet: 10

Num_residues_per_peptide: 8

Reference_segment: 2-8

FibrilGen parameters

N: 50

theta_z: 5.0

Sign_theta_y: 1

Output_filename: F8_rod.pdb

Show_the_structure

- Parameter N is an integer repeating stacking along the vertical axis.
- Parameter θ_z is the initial tilt angle.
- Parameter "sign of θ_y"" assigned to 1 for left-handed twist, and -1 for right handed twist.

Show code

Practice (class 3 **rod** structures)

1. Run the example
2. Try a different *N*. Increase the number for a longer fibril.
3. Try a different sign of θ_y. Assign 1 for a left-handed twist and -1 for a right-handed twist.
4. Try a different θ_z. Increase the number for a shorter fibril.



Interface (class 4 **stacked rod** structures)

(d) We use our optimised bilayer to assemble a **stacked rod** structure. We change the number of stacking on the fibril cross-section (parameter **K**, do not specify element 1 surrounded by element 0); the twist chirality (the sign of parameter θ_y), the initial tilt angle (parameter θ_z), and the number of stacking (parameter N) along the fibril axis.

[ ]

Reference bilayer

FibrilGen parameters

Input_filename	Input_num_peptides_per_sheet	Num_residues_per_peptide
F8_bilayer.pdb	10	8
Reference_segment	**N**	**K**
2-8	10	0_1,1_1
theta_z	**Sign_theta_y**	**Output_filename**
10.0	1	F8_stacked_rod.pdb

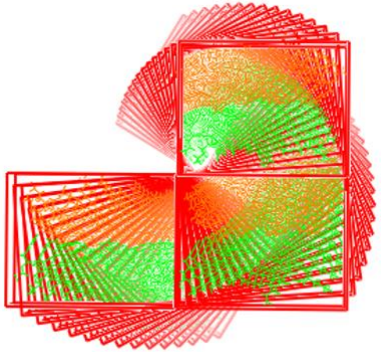
☑ Show_the_structure

- Parameter N is an integer repeating stacking along the vertical axis.
- Parameter **K** is a binary matrix ("_" to seperate columns, "," to sperate rows) assigning whether a 2 x 2 unit is presented.
- Parameter θ_z is the initial tilt angle.
- Parameter "sign of θ_y"" assigned to 1 for left-handed twist, and -1 for right handed twist.

Show code

Practice (class 4 **stacked rod** structures)

1. Run the example
2. Try a different *N*. Increase the number for a longer fibril.
3. Try a different **K**. Increase the number of presented units for a thicker fibril.
4. Try a different sign of θ_y. Assign 1 for a left-handed twist and -1 for a right-handed twist.
5. Try a different θ_z. Increase the number for a shorter fibril.



Interface (class 5 **ribbon** structures)

(e) We use our optimised bilayer to assemble a **ribbon** structure. We change the twist chirality (the sign of parameter θ_y), the initial tilt angle (parameter θ_z), the initial radius of rotation (parameter r_y), and the number of stacking (parameter N) along the fibril axis.

[ ]

Reference bilayer

FibrilGen parameters

Input_filename	Input_num_peptides_per_sheet	Num_residues_per_peptide
F8_bilayer.pdb	10	8
Reference_segment	**N**	**theta_z**
2-8	50	30.0
r_y	**Sign_theta_y**	**Output_filename**
60.0	-1	F8_ribbon.pdb

☑ Show_the_structure

- Parameter N is an integer repeating stacking along the vertical axis.
- Parameter θ_z is the initial tilt angle.
- Parameter r_y is the initial radius of rotation.
- Parameter "sign of θ_y"" assigned to 1 for left-handed twist, and -1 for right handed twist.

Show code

Practice (class 5 **ribbon** structures)

1. Run the example
2. Try a different *N*. Increase the number for a longer fibril.
3. Try a different sign of θ_y. Assign 1 for a left-handed twist and -1 for a right-handed twist.
4. Try a different θ_z. Increase the number for a shorter fibril.
5. Try a different r_y. Increase the number for a wider fibril.



Interface (class 6 **stacked ribbon** structures)

(f) We use our optimised bilayer to assemble a **stacked ribbon** structure. We change the number of stacking (parameter M) and the angle for stacking (parameter θ_s) on the fibril cross-section; the twist chirality (the sign of parameter θ_y), the initial tilt angle (parameter θ_z), the initial radius of rotation (parameter r_y), and the number of stacking (parameter N) along the fibril axis.

Reference bilayer

FibrilGen parameters

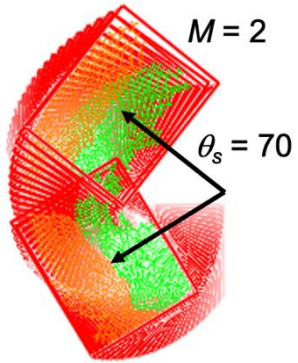
Input_filename: F8_bilayer.pdb	Input_num_peptides_per_sheet: 10	Num_residues_per_peptide: 8
Reference_segment: 2-8	N: 20	theta_s: 120.0
M: 2	theta_z: 10.0	r_y: 60.0
Sign_theta_y: 1	Output_filename: F8_stacked_ribbon.pdb	Show_the_structure

- Parameter N is an integer repeating stacking along the vertical axis.
- Parameter θ_s is angular displacement of units on the fibril cross-section.
- Parameter M is the number of stacking on the fibril cross-section.
- Parameter θ_z is the initial tilt angle.
- Parameter r_y is the initial radius of rotation.
- Parameter "sign of θ_y"" assigned to 1 for left-handed twist, and -1 for right handed twist.

Show code

Practice (class 6 **stacked ribbon** structures)

1. Run the example
2. Try a different *N*. Increase the number for a longer fibril.
3. Try a different sign of θ_y. Assign 1 for a left-handed twist and -1 for a right-handed twist.
4. Try a different θ_z. Increase the number for a shorter fibril.
5. Try a different r_y. Increase the number for a wider fibril.
6. Try a different *M*. Increase the number of presented units for a more enclosed fibril.
7. Try a different θ_s. Increase the number for a more twisted fibril.



Enjoy building cross-beta fibrils?

We welcome feedback and suggestions!